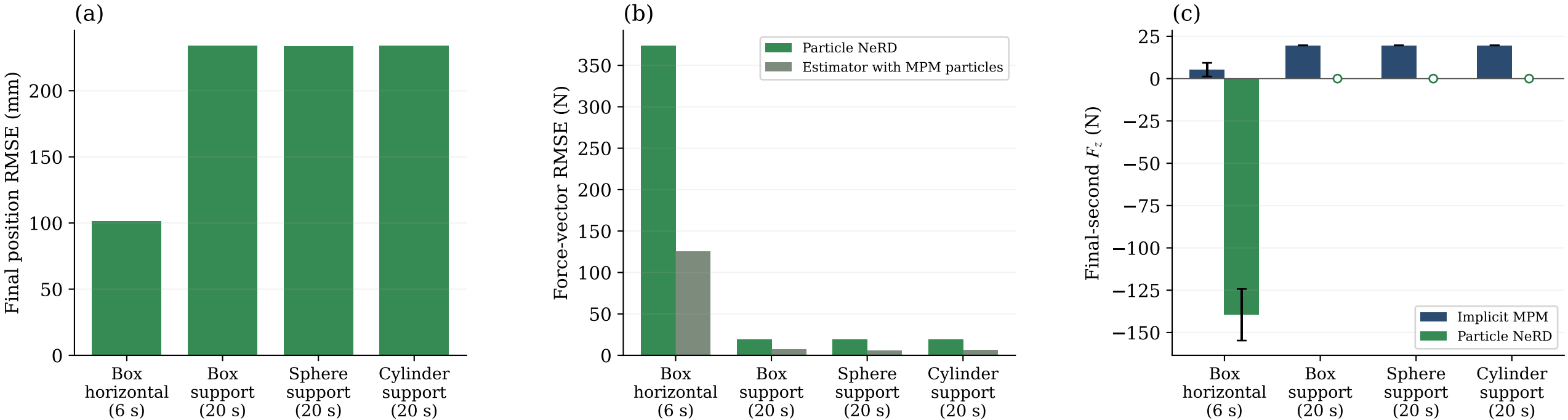

Figure 7: Frozen object-model checkpoint 9,000. (a) Final particle-position RMSE; (b) force-vector RMSE without division by three; (c) mean Fz during the final second, with population standard deviation over time. Open circles mark measured zeros. Whiskers in (c) are temporal variation in one recording, not repeated-run uncertainty. The horizontal case lasts 6 s; support cases last 20 s. Body poses are replayed from MPM and do not respond to estimated force.